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Offline Reinforcement Learning for a Small Language Model in Reference Games. A Reproduction Study Using clembench

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Link to GitHub Repository

Motivation

- RL is increasingly used to fine-tune LLMs for **better generalisation** compared to SFT.
- Gül & Artzi (2024) reported significant improvements via **joint inference** (generation + comprehension) in a **ReferenceGame (image based)** setup.
- Reproduction and analyses of those results using open-source tools to test **reproducibility** and **human influence** in quality of fine-tuning LLMs.

Research Questions

i. Can the improvements reported by Gül & Artzi be reproduced?
ii. What is the relative contribution of **human interaction** vs **algorithmic optimisation**?

Experimental Setup

The ReferenceGame (Dialogue based)

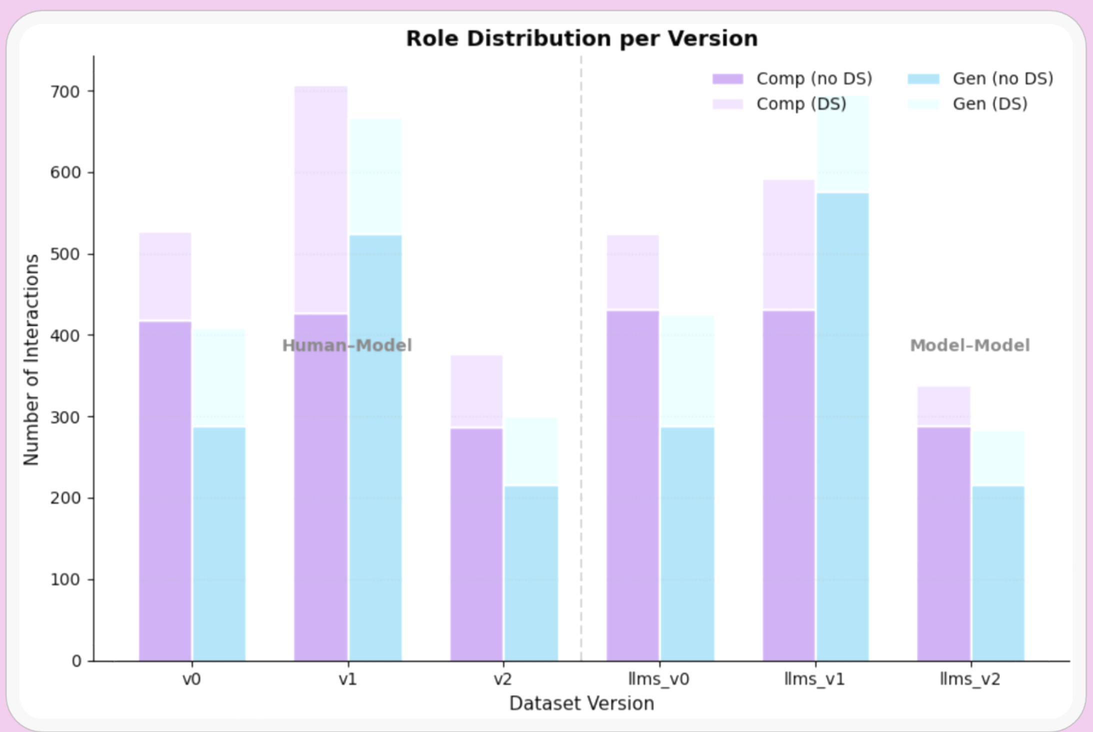
- 2-player, single-turn, collaborative task: **Describer & Guesser**
- 5×5 grids (letters, numbers, shapes, symbols) e.g.:

```
x x x x x    x x x x x    x x x x x    x x x x x    x □ □ □ x    x x x x x
x x □ x x    x □ □ □ □    x x □ □ □    x □ □ □ □    x x x x x    x □ □ □ x
x x x x x    x □ □ □ □    x x x x x    x x x □ □    x x x x x    x x x x x
x x □ x x    x □ □ □ □    x x □ □ □    x □ □ □ □    x x x x x    x x □ □ □
x x x x x    x x x x x    x x x x x    x □ □ □ □    x □ □ □ x    x x □ □ □
```

- Evaluation of comprehension and generation skills.

Data Collection

- Human & Model: 14 participants via Discord/Zoom
- Model & Model: Automated self-play (Baseline)
- Data Sharing
- Data generation → Offline RL fine-tuning → New inference data → *Iterative update* (*v0 → v1 → v2*)



Win Ratio:

Model	v0	v1	v2	llms_0	llms_1	llms_2
Gen	0.56	0.63	0.58	0.54	0.4	0.42
Comp	0.44	0.6	0.46	0.44	0.47	0.35

Comment: Base model was Llama3.1-8B-Instruct.

Methodology

- Objective: maximise reward $r \in \{-1, +1\}$ for correct guesses
- REINFORCE loss with IPS:

$$L_{REINFORCE+IPS} = -c_i r \log P_\theta (\hat{t}' \mid I, u)$$

- Early Stopping based on comprehension accuracy
- Usage of QLoRA and HF's Transformers library

Results

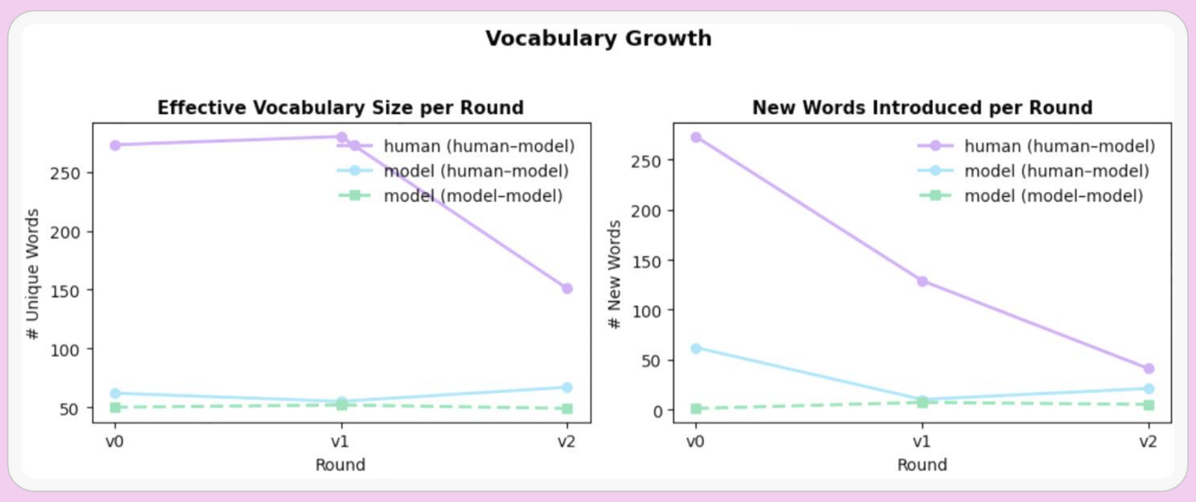
Findings from Clembench benchmark v3.0:

- RL fine-tuning did not replicate performance boost observed by Gül & Artzi.
- Human data improved linguistic richness but not task accuracy.
- Performance dropped in later versions, probably due to data imbalance.

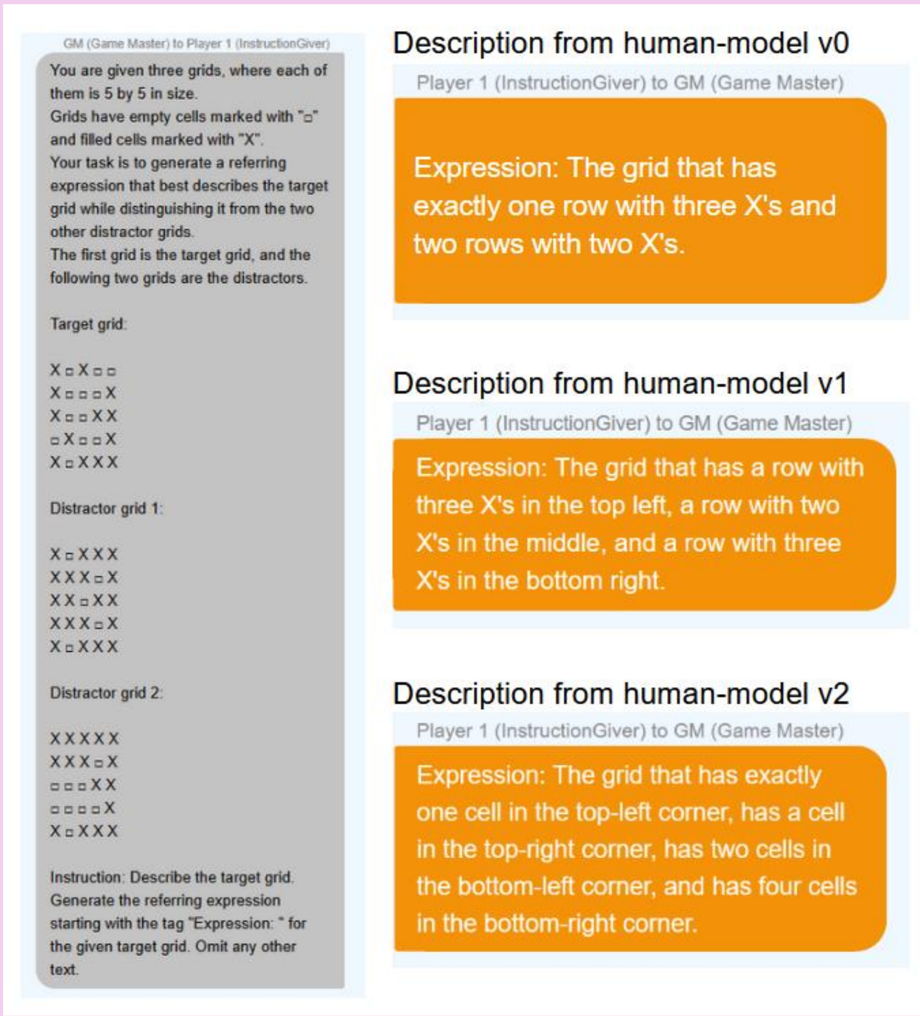
Model	ClemScore (Overall)	ClemScore (ReferenceGame)
v0	31.5	43.3
v1	29.5	40
v2	24.3	36.7
llms_v1	28.9	40
llms_v2	28.6	33.3

Language Analysis (Human & Model):

- *Vocabulary diversity* increased with human input, stabilising over time
- *Utterance lengths* converged between humans and models (~17–24 tokens).



- Common issues:
  - *Miscounting* elements
  - Vague or ambiguous phrasing
  - Lack of contrastive descriptions
- Human input improved *fluency*, but **not** precision or grounding.



Limitations

- Limited human participations and manual setup.
- GPU constraints prevented v3.
- Diverse toolchains not optimised for RL.

Future Work

- Browser-based Slurk interface (already done).
- Introduce generation-based early stopping criteria.
- Explore data quality and balance effects.

Conclusion

- Human input influences *how* models speak, but also *how well* they perform?
- RL for small models is promising.
- Reproducibility in RL-style pipelines was the core challenge due to hinderances by fast-evolving dependencies.

Acknowledgements

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References

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